

LETTER OPEN ACCESS

A Portable Intelligent Navigation System Based on Embedded Hybrid Computing for Visually Impaired People

Hongyi Zhang¹  | Tianyu Hu¹  | Mengran Zhang¹  | Lei Mo²  | Kunpeng Li³

¹School of Future Technology, Southeast University, Nanjing, China | ²Key Laboratory of Measurement and Control of CSE, Ministry of Education, School of Automation, Southeast University, Nanjing, China | ³Zhongda Hospital, Southeast University, Nanjing, China

Correspondence: Lei Mo (lmo@seu.edu.cn)

Received: 30 October 2025 | **Revised:** 17 January 2026 | **Accepted:** 26 January 2026

ABSTRACT

This paper presents a portable intelligent navigation system for visually impaired persons that integrates environmental perception, voice interaction, and real-time navigation, developed on an STM32 multicore embedded computing platform. The designed system acquires environmental information and users' commands via integrated modules, for example, camera, GPS, and microphone. Due to the limited data-processing capabilities of the STM32 platform, lightweight tasks are executed locally. In contrast, computation-intensive tasks, for example, image processing and path planning, are sent to the cloud for processing to enhance the system's real-time responsiveness. Experimental results demonstrate that the designed system can effectively guide visually impaired people to the desired destinations, which can achieve an accuracy of up to 90% in traffic element recognition, for example, traffic lights and crosswalks, an average response time of less than 5 s, and a navigation accuracy of ± 2 m.

1 | Introduction

With the global trend of aging and the increasing prevalence of chronic diseases [1], visual impairment has become a serious problem [2]. According to the report of the World Health Organization (WHO), in 2019, approximately 295 million people worldwide suffered from Moderate to Severe Visual Impairment (MSVI) problems, including 43.3 million who are completely blind [3]. Safe and efficient daily trips for visually impaired persons remain a challenging issue. Traditional assistance methods, such as blind canes and guide dogs, offer limited support, particularly in complex and dynamic urban environments, due to their restricted capacity for critical information exchange and feedback [4].

Recently, with the rapid development of Artificial Intelligence (AI), embedded/edge computing, and sensor technology [5], more efficient and sophisticated systems have been developed. For example, a wearable LiDAR device is developed in [6] with voice-guided navigation and vibration feedback. In [7], an

arm-based wearable device is designed for obstacle detection, navigation, and path planning. A smart cane integrating LiDAR and YOLOv5 for SLAM and obstacle detection is realised in [8]. However, the above systems primarily focus on onboard computation [9], thereby limiting their efficiency and scalability in real-world complex applications.

Compared with existing works, we combine cloud-based data processing with local embedded computation. Specifically, computationally intensive tasks such as image recognition [10] and path planning are offloaded to the cloud for processing, enabling high-accuracy perception and navigation services. Meanwhile, the STM32 embedded platform handles lightweight tasks such as real-time feedback and voice interaction [11]. This hybrid computing architecture optimises resource usage in the navigation system, improving responsiveness and energy efficiency of services in dynamic environments.

The designed intelligent navigation system integrates an STM32 embedded platform with multiple sensors, including the GPS,

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2026 The Author(s). *Electronics Letters* published by John Wiley & Sons Ltd on behalf of The Institution of Engineering and Technology.

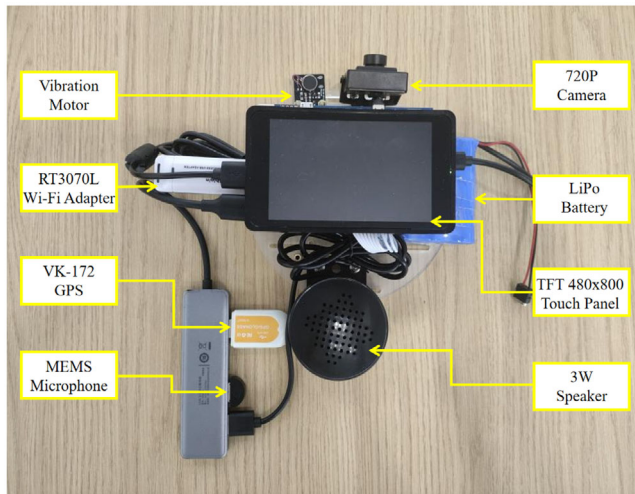


FIGURE 1 | Sensing, computing, communication, and interaction modules of the navigation system.

camera, microphone, and speaker, to enable environmental awareness and human-machine interaction, as shown in Figure 1. Specifically, STM32 and GPS are combined to acquire geographic location information, enabling real-time navigation services. During navigation, the STM32 connected to the camera to perform environmental awareness, capture real-time images of the surroundings, and support subsequent autonomous obstacle-avoidance decisions. The camera is oriented towards the user's walking direction and can be manually adjusted to observe a specific area. Furthermore, the STM32 accesses a microphone and a speaker, enabling voice interaction between the system and the user. Due to the low cost and low energy consumption of the designed system, we can achieve an energy-efficient data processing for environmental awareness, voice interaction, and navigation.

To provide flexible computation and account for the computational resources of the STM32 platform, the system adopts a hybrid computing model that combines onboard and cloud computing to balance computational efficiency and real-time response in practical applications. The STM32 system serves as an energy-efficient process unit, during the onboard computing phase, primarily responsible for real-time sensor data acquisition and preliminary data processing, for example, acquires raw image data from the camera and converts the navigation instructions from text to voice. On the other hand, the system leverages the high-performance computing capabilities of the cloud to complete complex, computationally intensive tasks. Specifically, the cloud (e.g., Alibaba Cloud) performs object identification (e.g., traffic lights and crosswalks) based on camera image data, determines the navigation decision (e.g., path planning and movement) through the user's position and requirement.

Compared to existing approaches, our main contributions can be summarised as follows.

- The designed system integrates STM32, camera, GPS, speaker, and microphone modules to enable multiple functions, including environmental awareness, voice interaction, and real-time navigation.

- The onboard + cloud computation architecture ensures the system's flexibility and responsiveness for real-time tasks, as well as the reliable operation of complex intelligent algorithms.
- We conducted a series of real experiments to evaluate the effectiveness and the performance of our navigation system. The experimental results demonstrate that our system provides efficient path planning and navigation through real-time environment awareness and obstacle avoidance.

In the following sections, we first introduce the system's hardware and software designs, then present the experimental results, and finally conclude this study.

2 | Embedded System Design and Implementation

The proposed intelligent navigation system is implemented on a hardware-software co-designed embedded platform that balances system performance, power consumption, and user requirements to ensure reliable services for visually impaired users in dynamic environments. The system mainly comprises four integrated components: (1) a hybrid computing framework for task allocation and closed-loop control, (2) a modular hardware architecture for sensing and actuation, (3) a power management subsystem for efficient energy distribution, and (4) a mechanical design for compactness and robustness.

2.1 | Hybrid Computing Framework

We implement the navigation system on an STM32MP157F platform [12], which features a dual-core heterogeneous architecture with an 800 MHz 32-bit Cortex-A7 for high-performance tasks (e.g., cloud communication and image processing) and a 200 MHz 32-bit Cortex-M4 for real-time and energy-efficient tasks (e.g., sensor data collection and actuator output control).

The designed system adopts a fixed task partitioning, based on the task characteristics and the computational resources of the local device and the cloud. During the hybrid data processing, sensor data are first preprocessed locally by the Cortex-A7 core to reduce communication overhead between the local device and the cloud. The high-level inference results are then obtained from the cloud API (Application Programming Interface), for example, Alibaba Qwen-VL, Gaode (AMAP), and Baidu ASR (Automatic Speech Recognition) cloud services, and translated into real-time control commands executed on the embedded platform [13]. These commands can be used to drive the actuators, for example, speaker and vibration motor, forming a closed loop of sensing, computation, and actuation.

- For visual perception, the camera's raw image frames are first resized and converted from the native YUYV format to RGB, then compressed into JPEGs to minimise transmission bandwidth. The images are encoded in Base64 and encapsulated in JSON requests, which are transmitted via HTTPS to the cloud for scene understanding and image recognition. The returned results from the cloud, that is, textual responses such as traffic

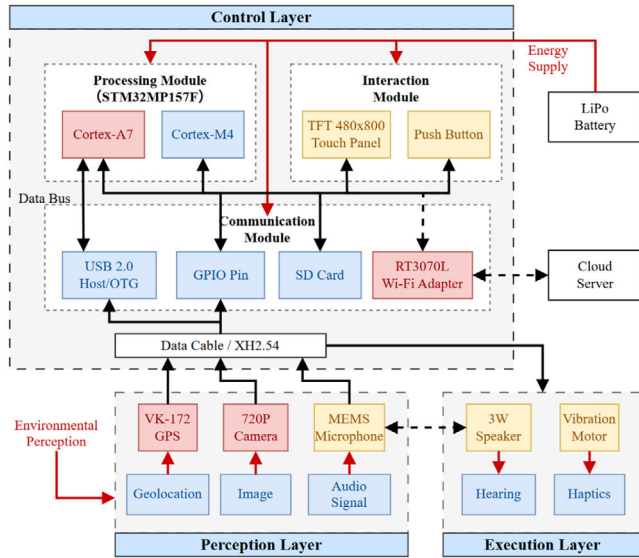


FIGURE 2 | Hardware architecture of navigation system.

lights, pedestrian crossings, and recommended actions, are subsequently converted into voice feedback to the user.

- For positioning, the raw NMEA-0183 data (e.g., time, latitude, longitude, and positioning status) from the GPS is first parsed by Cortex-A7 to extract valid position information. The GPS data are decoded, and the latitude and longitude values are converted from the NMEA degrees-minutes format to decimal degrees in the WGS-84 coordinate system. The coordinates, timestamped with UTC (Coordinated Universal Time), are then transmitted to the Gaode cloud for path planning.
- For auditory input, microphone data are recorded as 16-bit, 16 kHz PCM (Pulse Code Modulation) audio and stored as WAV files. The Cortex-A7 performs basic format and integrity checks before uploading the raw audio stream to the Baidu ASR cloud via HTTP POST for remote decoding and recognition. The recognised text returned in JSON format is then used to interpret user speech commands, enabling convenient human-machine interaction [14].

While awaiting data processing on the cloud, the local device remains active, performing tasks such as deviation detection and voice prompts, which improves overall efficiency. To guarantee data security, data transmissions between the embedded device and the cloud services are encrypted using HTTPS/TLS. Data collection and processing on the embedded device are mainly controlled by the user, and the generated information is not retained permanently.

3 | Hardware Architecture

As shown in Figure 2, the hardware of our intelligent navigation system comprises several modules to enhance scalability and maintainability.

- **Perception Layer:** Environmental perception is achieved by multiple sensors such as GPS, camera, and microphone, where the VK-172 GPS module provides the information

of geolocation with metre-level accuracy under open-sky conditions; the 720p USB camera captures the surrounding visual information in a real-time manner, while the MEMS (Micro-Electro-Mechanical System) microphone allows the user to communicate with the system, for example, input the destination of navigation. The above sensing devices deliver the collected data to the upper control layer for processing.

- **Execution Layer:** Multimodal feedback (i.e., vibration and sound) is provided through a miniature vibration motor and a 3 W speaker, delivering tactile and auditory cues. Together with a 480×800 TFT touch panel and physical push buttons, these modules form the hardware foundation of the navigation system's human-machine interaction, ensuring a convenient user experience for visually impaired users.
- **Control Layer:** The STM32 serves as the core controller, integrating a dual-core Cortex-A7 and Cortex-M4 processors for sensor data processing and execution task management. Peripheral connectivity is supported through the USB 2.0 Host/OTG (GPS, camera, microphone, speaker, and Wi-Fi adapter), GPIO pins (vibration motor), and SD card storage (Linux-based operating system). In addition to performing cloud computing, the STM32 connects to the Internet via an RT3070L Wi-Fi adapter. Most peripherals rely on native OpenSTLinux drivers to simplify integration, while non-USB peripherals, such as the vibration motor, employ XH2.54 standard connectors to ensure signal integrity.

3.1 | Power Management

The system is powered by a 5 V/3 A LiPo battery pack, providing 4–6 h of continuous operation under typical workloads, for example, destination navigation and object identification. The battery supplies power via a Type-C interface to the STM32MP157F platform, which integrates the STPMIC1 power management IC. The STPMIC1 converts the 5 V input into multiple regulated voltage rails for different functional domains: the 1.1 V rail powers the Cortex-A7/M4 cores and internal logic circuits, the 1.8 V rail supplies high-speed interfaces such as MIPI DSI (Display Serial Interface for touch panel) and SDMMC (Secure Digital/MultiMediaCard Interface for SD card), and the 3.3 V rail drives the vibration motor through GPIO pins and powers the touch panel via the DSI interface. Meanwhile, the 5 V input is directly distributed to the USB ports to power peripheral devices, for example, the GPS, camera, microphone, and Wi-Fi adapter. This integrated power architecture enhances overall stability, simplifies power distribution, and ensures efficient energy utilisation across the embedded platform.

3.2 | Mechanical Design

The navigation system is mechanically organised into three layers, forming a compact and stable structure, as shown in Figure 3. All hardware components are enclosed in a lightweight transparent acrylic case, which provides physical protection while maintaining portability.

- The top layer features a touchscreen display that provides real-time visualisation of navigation data and system status

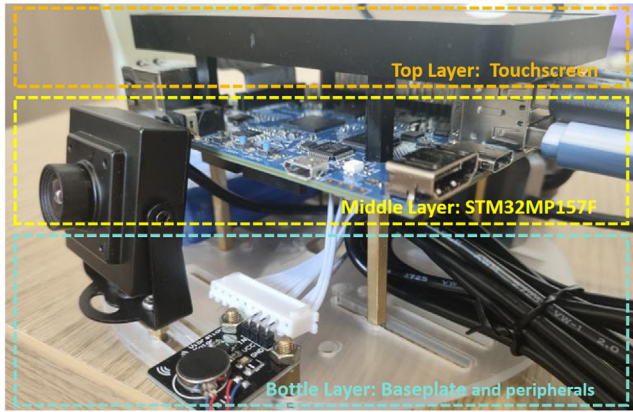


FIGURE 3 | Three-layer embedded computing platform design.

via a graphical user interface, enabling intuitive human-machine interaction.

- The middle layer contains the STM32MP157F development board, integrating the primary processing centers and peripheral interfaces, for example, USB 3.0, HDMI, Ethernet, and Type-C connectors.
- The bottom layer includes an acrylic baseplate that fixes peripherals: the camera and vibration motor at the front, the speaker, microphone, and GPS module at the rear, the Wi-Fi module on the left, and the battery compartment on the right.

This layered architecture isolates functional domains, ensures organised wiring, and reduces electromagnetic interference between modules. The copper standoffs provide both structural rigidity and passive heat dissipation for the main board. Overall, the mechanical design achieves an optimal balance between compactness, stability, and thermal performance, ensuring reliable operation. In addition, integrating physical buttons and voice-based communication supports intuitive interaction, thereby improving applicability in daily navigation scenarios.

4 | Software Architecture

The system software aims to provide rapid response, stable operation, and scalable functionality for users, which comprises two logical levels: the interaction layer and the decision-making layer, as shown in Figure 4.

4.1 | Interaction Layer

This layer provides the direct interaction between the system and the user through the GUI, microphone, vibration motor, and physical buttons. It captures the user's intent via speech input, provides system feedback through visual and tactile means, and enables intuitive navigation. As shown in Figure 5, the GUI is developed based on GTK (GIMP Toolkit), and it contains the following four parts:

- **Internet Connection:** This module manages the wireless network configuration of the devices and enables the terminal to access the Internet through the Wi-Fi adapter. By establishing

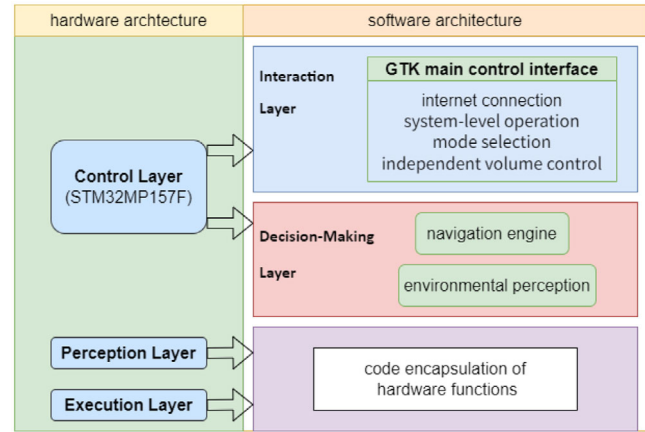


FIGURE 4 | Software architecture.

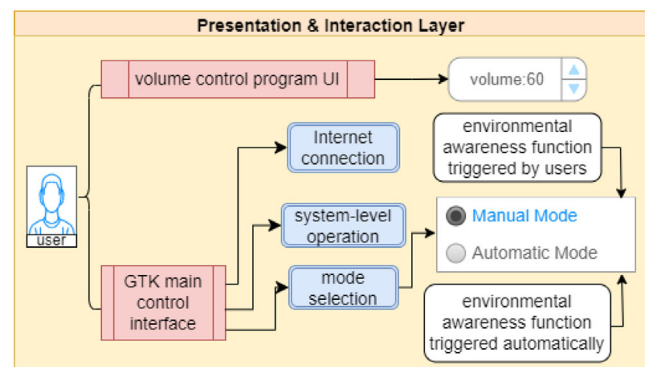


FIGURE 5 | GUI architecture.

a stable communication link, the navigation system can use collaborative edge-cloud computation.

- **System-Level Operation:** The interface provides graphical entries for core system operations such as exit, restart, and shutdown. Through an intuitive visual layout and one-click execution, users can manage system states efficiently without typing commands, enhancing operational convenience and safety.
- **Mode Selection:** After the network has been successfully initialised, the user can select a proper navigation mode (e.g., Mode 1 or Mode 2). Mode 1 is a manual mode in which the user triggers environmental awareness via physical buttons. While Mode 2 is an automatic mode, where the environmental awareness is performed automatically with a fixed time period (e.g., every 2 s), continuously broadcasting the road conditions ahead for the user. GUI can convert this high-level user choice into explicit behavioral commands and forward them to the underlying control modules. This mechanism ensures a clean separation between user interaction and execution logic while supporting quick mode switching in practical scenarios.
- **Independent Volume Control:** To avoid coupling with complex navigation tasks, the volume adjustment function is implemented as an independent lightweight process. This design guarantees the real-time responsiveness and high availability

of auditory feedback, ensuring that voice prompts remain clear and uninterrupted throughout system operation.

4.2 | Decision-Making Layer

This layer is the “brain” of the system, responsible for executing core logic tasks, such as path planning, real-time navigation, environmental sensing, and decision-making. It is based on two distinct modules (navigation engine and environmental perception) to ensure timely guidance across different scenarios, with the perception and planning pipeline updated at approximately 200–500 ms.

4.2.1 | Navigation Engine

The navigation engine guides users from their current locations to their desired destinations, with internal operating logic that comprises two fundamental stages: path planning and real-time positioning.

- *Path Planning:* The navigation begins with the voice input of the user for the destination through the microphone. The vocal command is promptly converted into text through the online ASR service (i.e., Baidu ASR service). The resulting text is then used to construct a web request. This request is sent to the Gaode cloud service and then returned as JSON path information. Then, the system parses this data, extracting the latitude and longitude coordinates of all critical points of paths. These coordinates are formatted into an internal 2D array, effectively creating a digital map that serves as the foundation for the entire navigation process.
- *Real-Time Positioning:* Once the path is established, the system performs the real-time position and yaw detection. This stage runs on Cortex-A7, ensuring maximum responsiveness even without a network connection. The navigation engine continuously retrieves current geographic coordinates from the GPS module, typically with 1 s intervals. Haversine Spherical Distance Formula (HSDf) is used to calculate the distance between the current position and each array point along the established path, enabling precise determination of the user's location along the planned route.

To mitigate potential GPS signal drift, the system employs a yaw-detection mechanism based on a sliding window [15]. If the user's position consistently deviates from the predefined path, the route will be replanned using the Gaode API. As the user walks along the planned path, the system generates voice commands and activates the vibration motor to remind the user of further movements, such as turning left/right or stopping.

4.2.2 | Environmental Perception

The environmental perception aims to provide users with a real-time understanding of the physical world around them. It relies on a large language model (Qwen-VL) in the cloud to complete complex tasks, such as semantic image understanding



FIGURE 6 | Navigation and path planning.

and transforming visual information into speech descriptions that users can understand.

The environmental perception method includes the following steps: (1) image acquisition, (2) encoding and requesting, (3) cloud analysis, and (4) analysis and broadcasting. In the first step, the system captures an image of the scene in front of the user through the camera. Then, the system calls the API of the “Tongyi Qwen” large-scale visual language model (Qwen-VL) on Alibaba Cloud. The cloud service analyses the image and returns a descriptive text summary. To effectively convert the captured images into corresponding descriptive texts, we deeply customise the prompt (prompt words) of the model to make it focus on (1) the road direction, (2) the existence of obstacles in the front, and (3) the status of traffic lights. To improve real-time response and ensure the voice broadcast is clear and concise, the output text is limited to 50 words. For example, the implemented prompt in our case is set to: Describe the road ahead for a visually impaired person. Include road direction, nearby ground obstacles, and traffic light status if present. Respond within 50 Chinese characters using only Chinese text with commas or periods. This design ensures the model focuses on relevant visual cues and produces consistent, concise outputs suitable for real-time navigation assistance.

5 | Experimental Result Analysis

To verify the navigation performance and obstacle-avoidance capabilities of the system designed in this paper, especially in complex urban environments, we conduct several experiments based on path planning and image recognition. The experiment selected a real road near Southeast University in Nanjing, China, as the test scenario, which includes multiple traffic lights and pedestrian crossing lines. The experimental results are summarised as follows:

- *Navigation and Path Planning:* Figure 6 shows the navigation results, where the red and green points represent the start and end points (destination) set by the user, respectively. The length of the planned route is approximately 1900 m, as shown by the blue line. At the beginning, the system will perform real-time positioning. Due to complex environments, such as high-rise buildings, there are position errors. To improve accuracy, we can perform manual calibration before navigation begins. Specifically, we select several landmarks (e.g., the south and west gates of Southeast University), conduct corresponding tests at these locations, and observe a consistent systematic error between the device's reported coordinates and the actual ground-truth positions. Then, we

use the errors to calibrate the original GPS readings. In our system, we set $lat_{calibrated} = lat_{raw} + \Delta lat$, where $lat_{calibrated}$ and lat_{raw} are the adjusted and measured latitude values, respectively, and Δlat is the offset, set to -0.002092° based on experience. The position error is controlled within ± 2 m through the calibration.

After getting the start point, the user can provide the destination through voice input. During the experiments, we use the default path provided by Gaode cloud service, which is usually the nearest path. The navigation response delay, measured by the time from the navigation-related event is triggered (e.g., route deviation or voice command) to the information (e.g., audio or vibration) feedback to the user, is on average less than 5 s. During the navigation process, when the user deviates from the planned route, for example, the user moves along with the red arrow, the system would give a voice prompt, until the user returns to the planned route again. In the experiments, the voice prompt is triggered when the deviation reaches 10 m. If the user did not return to the correct path within the specified time, the system would replan the path.

- **Image Recognition:** During navigation and path planning, the user should be aware of the surrounding environment. For example, when pedestrians are in front of the user, the system should prompt the user to perform obstacle avoidance; when the user crosses the intersection, the system should prompt the user to stop or continue, depending on the traffic light states. In the designed system, the user can select Mode 1 or Mode 2 to perform environmental awareness. During the experiments, we select Mode 2, that is, automatic mode with 2 s sample interval for broadcasting road conditions. In this mode, the response latency, defined by the time from image recognition to providing the audio feedback, is typically below 1.6 s, ensuring timely guidance for users.

The designed system has been evaluated under diverse outdoor environments (e.g., overcast weather and bright sunlight) and typical urban scenarios (e.g., urban streets and road intersections). Experimental results show that the key traffic elements (e.g., traffic lights and pedestrian crossings) can be recognised with an average accuracy of approximately 90%, demonstrating robust performance across different real-world scenarios. (1) Traffic lights: a correct recognition requires not only detecting the presence of a traffic light but also accurately classifying its current state as either red or green. (2) Pedestrian crossings: a correct recognition requires detecting the existence of a pedestrian crossing at the intersection. The accuracy of each category (traffic lights and pedestrian crossings) is calculated by $\text{Accuracy} = (\text{Number of correctly recognised events} / \text{Total number of recognised events}) \times 100\%$, where an event represents both existence detection and state recognition. Figure 7 shows the results of traffic light and pedestrian crossing recognition.

6 | CONCLUSION

This article presents a low-cost, highly portable navigation system for visually impaired persons that provides walking assistance. The designed system is based on an STM32 multicore embedded platform that integrates multiple hardware modules, including the camera, microphone, vibration motor, speaker, GPS, and Wi-



FIGURE 7 | Traffic light and pedestrian crossing recognition.

Fi. In addition, our system uses an onboard and cloud-based computing framework to enhance its real-time responsiveness and handle complex tasks, such as path planning, image recognition, and real-time navigation. A series of real experiments is conducted to evaluate the performance of our systems. The results show that the designed system can provide path planning and obstacle-avoidance functions, with accurate positioning and image recognition, including traffic lights and pedestrian crossings. In the future, we plan to reduce the device's size and expand the range and scope of image recognition to support applications in more complex environments.

Author Contributions

Hongyi Zhang: data curation. **Tianyu Hu:** methodology. **Mengran Zhang:** data curation. **Lei Mo:** conceptualization. **Kunpeng Li:** conceptualisation.

Acknowledgements

Hongyi Zhang, Tianyu Hu, and Mengran Zhang contributed equally to this work. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFF0902800, in part by the Natural Science Foundation of Jiangsu Province under Grant BK20242028 and Science and Technology Planning Project of Guangdong Province under Grant 2025B0101080001.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data available on request from the authors.

References

1. J. Chen, Y. Zhu, Z. Li, et al., "Global Impact of Population Aging on Vision Loss Prevalence: A Population-Based Study," *Global Transitions* 6 (2024): 28–36.
2. L. Que, Q. Zhu, C. Jiang, and Q. Lu, "An Analysis of the Global, Regional, and National Burden of Blindness and Vision Loss Between 1990 and 2021: The Findings of the Global Burden of Disease Study 2021," *Frontiers in Public Health* 13 (2025): 1–10.
3. D. Abdulhussein and M. A. Hussein, "WHO Vision 2020: Have We Done It?" *Ophthalmic Epidemiology* 30, no. 4 (2023): 331–339.
4. M. Hersh, "Wearable Travel Aids for Blind and Partially Sighted People: A Review With a Focus on Design Issues," *Sensors* 22 (2022): 1–44.
5. M. M. Islam, M. Sheikh Sadi, K. Z. Zamli, and M. M. Ahmed, "Developing Walking Assistants for Visually Impaired People: A Review," *IEEE Sensors Journal* 19, no. 8 (2019): 2814–2828.

6. P. Chitra, V. Balamurugan, M. Sumathi, N. Mathan, K. Srilatha, and R. Narmadha, "Voice Navigation Based Guiding Device for Visually Impaired People," *International Conference on Artificial Intelligence and Smart Systems* (2021), 911–915.
7. Z. Li, F. Han, and K. Zheng, "An RGB-D Camera-Based Wearable Device for Visually Impaired People: Enhanced Navigation With Reduced Social Stigma," *Electronics* 14, no. 11 (2025): 1–20.
8. C. Mai, H. Chen, L. Zeng, et al., "A Smart Cane Based on 2D LiDAR and RGB-D Camera Sensor-Realizing Navigation and Obstacle Recognition," *Sensors* 24, no. 3 (2024): 1–23.
9. G. I. Okolo, T. Althobaiti, and N. Ramzan, "Assistive Systems for Visually Impaired Persons: Challenges and Opportunities for Navigation Assistance," *Sensors* 24, no. 11 (2024): 1–15.
10. T. Azzino, M. Mezzavilla, S. Rangan, Y. Wang, and J.-R. Rizzo, "5G Edge Vision: Wearable Assistive Technology for People With Blindness and Low Vision," *IEEE Wireless Communications and Networking Conference* (2024), 1–6.
11. X. Ji, F. Gong, N. Wang, J. Xu, and X. Yan, "Cloud-Edge Collaborative Service Architecture With Large-Tiny Models Based on Deep Reinforcement Learning," *IEEE Transactions on Cloud Computing* 13, no. 01 (2025): 288–302.
12. STMicroelectronics, "STM32MP157F Series Microprocessors," <https://www.st.com/en/microcontrollers-microprocessors/stm32mp157f.html> (2024).
13. Y. Wang, C. Yang, S. Lan, L. Zhu, and Y. Zhang, "End-Edge-Cloud Collaborative Computing for Deep Learning: A Comprehensive Survey," *IEEE Communications Surveys & Tutorials* 26, no. 4 (2024): 2647–2683.
14. C. Deuerlein, M. Langer, J. Seßner, P. Heß, and J. Franke, "Human-Robot-Interaction Using Cloud-based Speech Recognition Systems," *Procedia CIRP* 97 (2021): 130–135.
15. W. Liu, H. Zhang, L. Ma, and Z. Deng, "Inertial Navigation With Adaptive Sliding Window and Spatiotemporal Network," *IEEE Sensors Journal* 25, no. 10 (2025): 18432–18439.